

Assignment_1_Answer Key

1. (a) $25\sqrt{2} \cos(\omega t - 140^\circ) V$
(b) $13\sqrt{2} \cos(\omega t + 67.38^\circ) A$
(c) $5\sqrt{2} \cos(\omega t + 126.87^\circ) A$
(d) $8\sqrt{2} \cos(\omega t + 70^\circ) V$
2. (a) V_2 leads V_1 by 30°
3. (a) $6.91 \angle 12.81^\circ$ (b) $0.565 \angle -160.13^\circ$ (c) $-5.5 - j17.67$ (d) $8.293 + j7.2$
4. $(149.52 - j195)\Omega$
5. $V_0 = 35.36 \cos(10t - 105^\circ) V$
6. $V_1 = 11.325 \cos(2t + 60.01^\circ) V$ and $V_2 = 33.02 \cos(2t + 57.12^\circ) V$
7. $Z_{th} = (12.4 - j3.2) \Omega$ and $V_{th} = 63.24 \angle -51.57^\circ V$
8. $1.465 \angle 38.48^\circ A$
9. $[3.5 + 5.445 \cos(20t - 10^\circ)] kW$ and $3.5 kW$
10. $Z_L = (3.415 - j0.7317) \Omega$ and $P_{max} = 51.47 W$
11. 0.8321 lag and $710 \angle 33.69^\circ VA$
12. (a) $44 \angle 70^\circ VA$, $44 VA$
(b) $15.05 W$, $41.35 VAR$
(c) 0.342 lag , $(94.06 + 258.4) \Omega$
13. 30.69 mF